

Day-12- Linux Networking Commands & Troubleshooting

Name: Perehan Sewid

THEORY

1. What is a network interface?

It is a hardware or a virtual connection that helps the computer to communicate with a network. and its names differ per machine and if the network is wired or wireless

2. What is the loopback interface?

'Lo', its usually 127.0.0.1 and it helps us to test tcp/ip stack without connecting to a network

3. What does ip addr display?

It shows network interfaces and their addresses.

4. What is the purpose of ip link?

It displays and manages network interfaces and shows their link-layer information.

5. What does hostname -I display?

It displays ip addresses assigned to the system.

6. What does ip route show?

It shows the routing table , it basically tells linux where to send the traffic if it doesn't know

7. What is a default route?

The default gateway is where you have to send the traffic before going to outside your local network.

8. What does ping test?

It tests whether a device is reachable on network, however a failed ping does not mean the device is offline maybe ICMP is blocked

9. Why can a failed ping be misleading?

Because maybe icmp is blocked on a device and we think the device is offline

10. What is DNS?

It is domain name server , its a protocol that translates domain names to ip addresses

11. What does getent hosts do?

It displays ip address related to a hostname.

12. What information does ip neigh display?

It displays ip, mac addresses to nearby devices and their network status.

13. What is a socket?

A socket is an endpoint used for network communication between devices or programs.

14. What does ss -tuln display?

Ss shows socket state

-t shows tcp sockets

-u shows udp sockets

-l shows listening ports

-n shows addresses and ports numbers numerically

15. What is the difference between TCP and UDP conceptually?

Tcp is slower but reliable

Udp is faster and connectionless

16. What is curl used for?

It is used to send and receive data commonly on websites to test urls using http or https

17. What does curl -I request?

It requests the http header ;content type ,server and status

18. What is traceroute used for?

It is used to show the route of a package when traveling to a destination

19. Why is ss commonly preferred over netstat on modern Linux?

Because it's faster and more efficient and shows detailed information about network connections and sockets.

20. Why are Linux networking commands useful in cybersecurity?

They help security professionals monitor network connections, identify devices and services, troubleshoot networks, and detect unusual or suspicious activity.

PRACTICAL SCREENSHOTS AND TROUBLESHOOTING

For the trouble shooting question i will use getent hosts [google.com](https://www.google.com) to test dns resolution

Because the problem is most likely that the dns cannot resolve the hostname [google.com](https://www.google.com) to the ip 8.8.8.8

```
kali@kali: ~  
Session Actions Edit View Help  
  
(kali@kali)-[~]  
$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff  
    inet 192.168.1.9/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0  
        valid_lft 84456sec preferred_lft 84456sec  
    inet6 fe80::764b:4d5a:d385:61d2/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
  
(kali@kali)-[~]  
$ ip route  
default via 192.168.1.1 dev eth0 proto dhcp src 192.168.1.9 metric 100  
192.168.1.0/24 dev eth0 proto kernel scope link src 192.168.1.9 metric 100  
  
(kali@kali)-[~]  
$ ip link  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000  
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff  
  
(kali@kali)-[~]  
$ ping -c 4 127.0.0.1  
PING 127.0.0.1 (127.0.0.1) 56(84) bytes of data.  
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0.031 ms  
64 bytes from 127.0.0.1: icmp_seq=2 ttl=64 time=0.043 ms  
64 bytes from 127.0.0.1: icmp_seq=3 ttl=64 time=0.031 ms  
64 bytes from 127.0.0.1: icmp_seq=4 ttl=64 time=0.030 ms  
  
— 127.0.0.1 ping statistics —  
4 packets transmitted, 4 received, 0% packet loss, time 3059ms  
rtt min/avg/max/mdev = 0.030/0.033/0.043/0.005 ms  
  
(kali@kali)-[~]  
$ getent hosts google.com  
2a00:1450:4006:81c::200e google.com  
  
(kali@kali)-[~]  
$ ss -tuln  
Netid      State      Recv-Q     Send-Q     Local Address:Port     Peer Address:Port  
  
(kali@kali)-[~]  
$ curl -I https://example.com  
  
HTTP/2 200  
date: Wed, 23 Sep 2026 17:05:52 GMT  
content-type: text/html  
server: cloudflare  
last-modified: Tue, 22 Sep 2026 21:15:25 GMT  
allow: GET, HEAD
```